

How the **Blockchain** Can Add **Transparency** to **Healthcare**



Ever since the prices of bitcoin went sky rocketing, the computer world has been excited about the blockchain. Every day, new applications of the blockchain are being thought up, and the focus has shifted to using it in every conceivable human activity. This article explains how blockchain technology can be used in healthcare.

The blockchain is a distributed ledger to maintain records among users without relying on third parties. It exists as multiple copies spread over multiple nodes around the world. Everything is secure and each new block gets connected to the previous one. Also, the transactions stored in a block are immutable.

Let us suppose that we have ten participants in a blockchain network. Initially, everybody will be given an empty ledger. Now, if one person wants to carry out some transaction, i.e., store any new data on the blockchain, that person has to announce it to everybody else, so that every user can record that data on his/her personal ledger. Any data that gets on to the blockchain is immutable and distributed. After the ledger is full, it needs to be mined and connected to the main chain, after which the network (all ten participants) gets updates about the new block.

Hash function

To store the state of the blockchain, we use cryptographic hash functions. When any data is passed through this hash function, it generates a key to represent that data. This is a one-way function, which means you can go A->B but predicting B->A is almost impossible. Interestingly, we can represent the whole Internet as a single hash address.

For example, SHA256 (Secure Hash Algorithm) that is used as the

hash function) can store anything in 256 bits with almost no chance of collision. You can read more about this at <https://blockgeeks.com/guides/what-is-hashing3>.

Structure of a block

A block in the chain contains the block header, the hash pointer to the previous block header and the Merkle root. The hash pointer contains the address of the previous block and the hash of the data

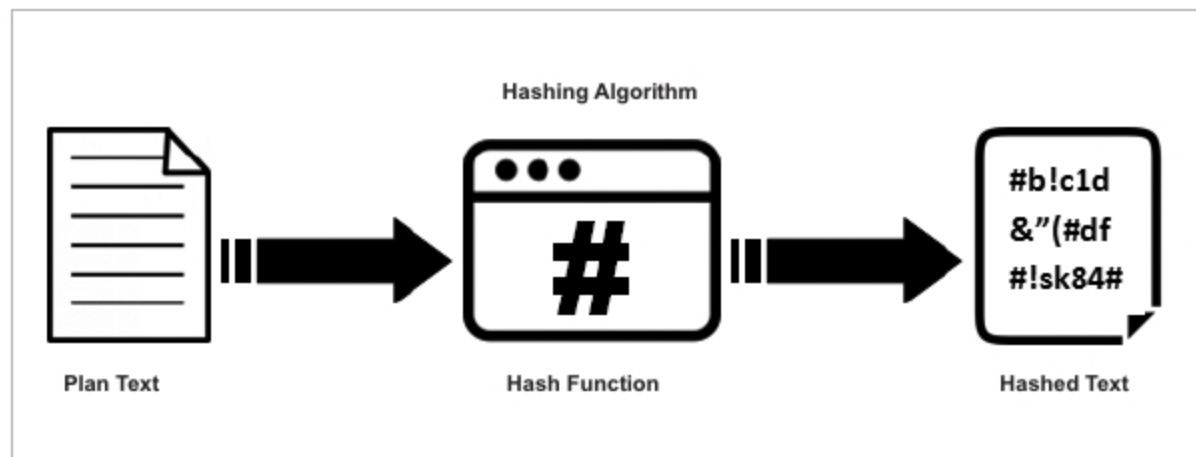


Figure 1: Hash function explained—it converts data to a hash, which is the same every time for the same input of data